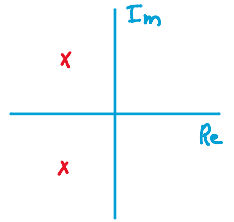


3 - Frequency response - Type 3 building blocks, nonminimum-phase systems & sketching procedure

Frequency response analysis

- 3rd class / Class 3: Second order complex conjugate pair of poles and zeros
- Frequency response of RHP zero
- Drawing rules bode plot



Class 3

$$\left[\underbrace{\left(\frac{j\omega}{\omega_n} \right)^2}_{\text{natural frequency}} + 2 \underbrace{\zeta}_{\text{Damping coefficient}} \frac{j\omega}{\omega_n} + 1 \right]^{\pm 1}$$

Break point $\frac{1}{\omega_n}$

Behaviour:

- $\omega/\omega_n \gg 1$, $\pm 40\text{dB/decade}$
- $\omega/\omega_n \approx 1$, resonant peak M_r
- Phase lead/lag 180° , 90° at ω_n

$$\frac{1}{\left(\frac{j\omega}{\omega_n} \right)^2 + 2\zeta \left(\frac{j\omega}{\omega_n} \right) + 1} \quad (\text{Complex conjugate pole pair})$$

$$\text{Re: } \frac{1 - \left(\frac{\omega}{\omega_n} \right)^2}{\left(1 - \left(\frac{\omega}{\omega_n} \right)^2 \right)^2 + \left(2\zeta \left(\frac{\omega}{\omega_n} \right) \right)^2} \quad \text{Im: } \frac{-2\zeta \left(\frac{\omega}{\omega_n} \right)}{\left(1 - \left(\frac{\omega}{\omega_n} \right)^2 \right)^2 + \left(2\zeta \left(\frac{\omega}{\omega_n} \right) \right)^2}$$

Three ranges:

- 1) $\omega \ll \omega_n$
- 2) $\omega \gg \omega_n$
- 3) $\omega = \omega_n$

$$\omega \ll \omega_n$$

$\frac{\omega}{\omega_n} \rightarrow 0$ is very small

$$\text{Re: } \frac{1 - \left(\frac{\omega}{\omega_n} \right)^2}{\left(1 - \left(\frac{\omega}{\omega_n} \right)^2 \right)^2 + \left(2\zeta \left(\frac{\omega}{\omega_n} \right) \right)^2} \approx \frac{1}{1+0}$$

$$\text{Im: } \frac{-2\zeta \left(\frac{\omega}{\omega_n} \right)}{\left(1 - \left(\frac{\omega}{\omega_n} \right)^2 \right)^2 + \left(2\zeta \left(\frac{\omega}{\omega_n} \right) \right)^2} \approx \frac{0}{1+0}$$

$$\text{Gain: } \sqrt{1^2 + 0^2} = 1 = 0\text{dB}$$

$$\text{phase: } \arctan\left(\frac{0}{1}\right) = 0^\circ$$

$$\omega \gg \omega_n$$

$$1 - \left(\frac{\omega}{\omega_n}\right)^2 \approx -\left(\frac{\omega}{\omega_n}\right)^2$$

$$\text{Re: } \frac{1 - \left(\frac{\omega}{\omega_n}\right)^2}{\left(1 - \left(\frac{\omega}{\omega_n}\right)^2\right)^2 + \left(2\zeta\left(\frac{\omega}{\omega_n}\right)\right)^2} \approx \frac{-\left(\frac{\omega}{\omega_n}\right)^2}{\underbrace{\left(-\left(\frac{\omega}{\omega_n}\right)^2\right)^2 + \left(2\zeta\left(\frac{\omega}{\omega_n}\right)\right)^2}_{\text{small compared to}}} \approx -\frac{\left(\frac{\omega}{\omega_n}\right)^2}{\left(\frac{\omega}{\omega_n}\right)^2} = -\left(\frac{\omega}{\omega_n}\right)^{-2}$$

$$\text{Im: } \frac{-2\zeta\left(\frac{\omega}{\omega_n}\right)}{\left(1 - \left(\frac{\omega}{\omega_n}\right)^2\right)^2 + \left(2\zeta\left(\frac{\omega}{\omega_n}\right)\right)^2} \approx \frac{-2\zeta}{\left(\frac{\omega}{\omega_n}\right)^3}$$

$$\text{Gain: } \left[\left(\frac{1}{\omega/\omega_n}\right)^2 + \underbrace{\left(-2\zeta/\left(\omega/\omega_n\right)^3\right)^2}_{\text{small compared to}} \right]^{1/2} \approx \left(\frac{\omega}{\omega_n}\right)^{-4}^{1/2} = \left(\frac{\omega}{\omega_n}\right)^{-2}$$

$$\text{Phase: } \arctan\left(\frac{-2\zeta\left(\frac{\omega}{\omega_n}\right)^{-3}}{\left(\frac{\omega}{\omega_n}\right)^{-2}}\right) = \arctan\left(-2\zeta\left(\frac{\omega}{\omega_n}\right)^{-1}\right) = 0^\circ \text{ sign information ignored, } \pm 180^\circ$$

$$\text{In decibels: } 20\log_{10}\left(\frac{\omega}{\omega_n}\right)^{\pm 2} = \pm 2 \cdot 20\log_{10}\left(\omega/\omega_n\right)$$

$$\omega = \omega_n$$

$$\frac{\omega_0}{\omega_n} = 1$$

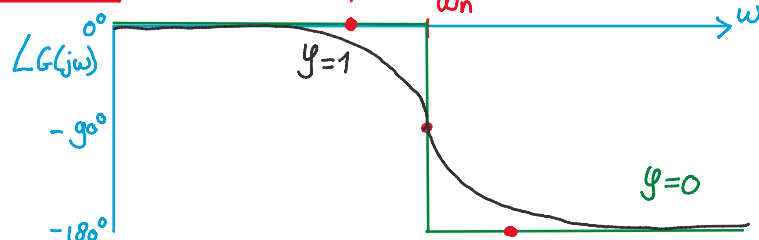
$$\text{Re: } \frac{1 - \left(\frac{\omega}{\omega_n}\right)^2}{\left(1 - \left(\frac{\omega}{\omega_n}\right)^2\right)^2 + \left(2\zeta\left(\frac{\omega}{\omega_n}\right)\right)^2} = 0$$

$$\text{Im: } \frac{-2\zeta\left(\frac{\omega}{\omega_n}\right)}{\left(1 - \left(\frac{\omega}{\omega_n}\right)^2\right)^2 + \left(2\zeta\left(\frac{\omega}{\omega_n}\right)\right)^2} = \frac{-1}{2\zeta}$$

	Zeros	Poles
$\zeta < 1/2$	-	+
$\zeta > 1/2$	+	-
+ : amplification		
- : attenuation		

$$\text{Phase: } \arctan\left(\frac{\text{Im}}{\text{Re}}\right) = \infty \text{ Very small}$$

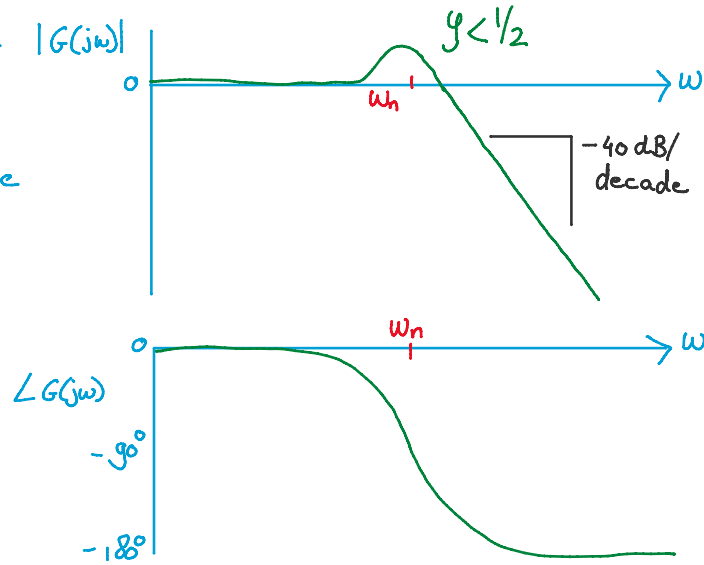
$$\text{Gain: } \sqrt{\left(\frac{-1}{2\zeta}\right)^2} = \frac{1}{2\zeta}, \text{ In decibels } \pm 20\log_{10}(2\zeta)$$



$$\begin{cases} \zeta = 1/2 & \log_{10}(2\zeta) = 0 \\ \zeta < 1/2 & \log_{10}(2\zeta) \text{ negative} \\ \zeta > 1/2 & \log_{10}(2\zeta) \text{ positive} \end{cases}$$

Class 3 bode plot

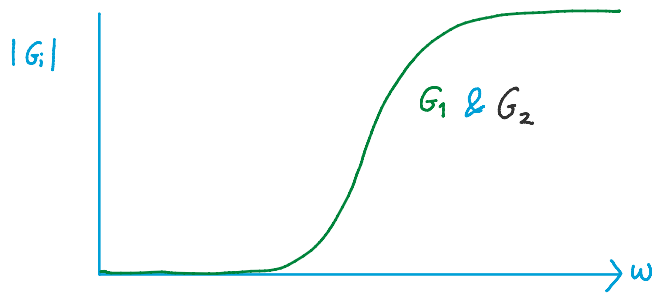
Complex conjugate pole pair



RHP zeros

$$G_1 = 10 \frac{s+1}{s+10}$$

$$G_2 = 10 \frac{s-1}{s+10}$$

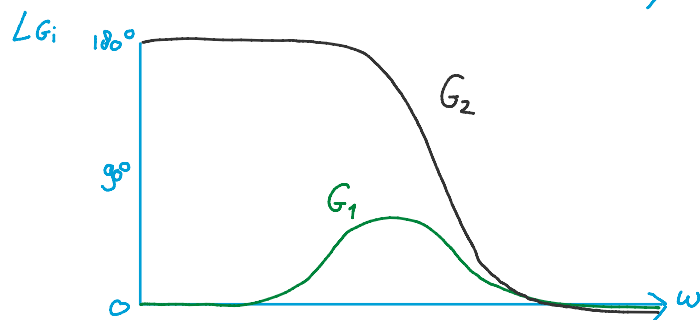


RHP zero

Gain: 20 dB/decade

Phase: -90°

$$|G_1(j\omega)| = |G_2(j\omega)|$$



Design procedure bode plots

- 1.) Write TF in bode form, identify building blocks and their break point frequencies
- 2.) Determine n in $K_o (j\omega_o)^n$, plot low Frequency asymptote through K_o
Bode gain-phase: gain $n \times 20$ dB/decade phase $n \times 90^\circ$
- 3.) Extend asymptote until first break point. Change slope based on pole/zero at break points
Continue in ascending order
- 4.) At the break points, magnitude deviates 3dB (1st order) or γ -dependent (2nd order)
- 5.) Plot low Frequency asymptote $n \times 90^\circ$
- 6.) Phase change of $\pm 90^\circ$ or $\pm 180^\circ$ at break points
- 7.) Draw asymptotes for phase and intersect them
- 8.) Account for 180° phase difference at asymptote intersection

Summary

- Frequency response of 2nd order system as Function of γ
$$\left[\left(\frac{j\omega}{\omega_n} \right)^2 + 2\gamma \left(\frac{j\omega}{\omega_n} \right) + 1 \right]^{\pm 1}$$
- Zero in LHP VS RHP
- Overall design procedure
Transfer function $\rightarrow$ Bode plot $\rightarrow$ System properties